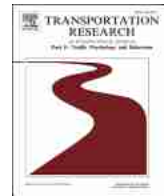




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Manual drivers' experience and driving behavior in repeated interactions with automated Level 3 vehicles in mixed traffic on the highway

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ABSTRACT

In the near future, conditionally automated vehicles (CAVs; SAE Level 3) will travel alongside manual drivers ($\leq$ SAE level 2) in mixed traffic on the highway. It is yet unclear how manual drivers will react to these vehicles beyond first contact when they interact repeatedly with multiple CAVs on longer highway sections or even during entire highway trips. In a driving simulator study, we investigated the subjective experience and behavioral reactions of $N = 51$ manual drivers aged 22 to 74 years ($M = 41.5$ years, $SD = 18.1$, 22 female) to driving in mixed traffic in repeated interactions with first-generation Level 3 vehicles on four highway sections (each 35 km long), each of which included three typical speed limits (80 km/h, 100 km/h, 130 km/h) on German highways. Moreover, the highway sections differed regarding the penetration rate of CAVs in mixed traffic (within-subjects factor; 0%, 25%, 50%, 75%). The drivers were assigned to one of three experimental groups, in which the CAVs differed regarding their external marking, (1) status eHMI, (2) no eHMI, and (3) a control group without information about the mixed traffic. After each highway section, drivers rated perceived safety, comfort, and perceived efficiency. Drivers were also asked to estimate the penetration rate of CAVs on the previous highway section. In addition, we analyzed drivers' average speed and their minimum time headways to lead vehicles for each speed zone (80 km/h, 100 km/h, 130 km/h) as well as the percentage of safety critical interactions with lead vehicles (< 1 s time headway). Results showed that manual drivers experienced driving in mixed traffic, on average, as more uncomfortable, less safe and less efficient than driving in manual traffic, but not as dangerous. A status eHMI helps manual drivers identify CAVs in mixed traffic, but the eHMI had no effect on manual drivers' subjective ratings or driving behavior. Starting at a level of 25% Level 3 vehicles in mixed traffic, participants' average speed decreased significantly. At the same time, the percentage of safety critical interactions with lead vehicles increased with an increasing penetration rate of CAVs. Accordingly, additional measures may be necessary in order to at least keep the existing safety level of driving on the highway.

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1. Introduction

The introduction of conditionally automated vehicles (CAVs) according to SAE Level 3 (SAE International, 2021) will create mixed traffic where manual drivers will have to interact with the CAVs instead of the passenger inside this CAV to resolve space-sharing conflicts on the highway (Markkula et al., 2020; Schieben et al., 2019). Typical space-sharing conflicts between two or more drivers on highways can occur when changing lanes or when accessing the highway (*merging paths* conflict), or when overtaking slower vehicles or passing broken-down vehicles (*obstructed paths* conflict; Markkula et al., 2020).

Since automated driving functions will first be introduced in the premium segment (see e.g., Audi, 2017; Daimler, 2019, 2021), it can be assumed that only a few vehicles will initially be equipped with Level 3 functions (see Kraftfahrtbundesamt, 2021). Beyond the introductory phase, the exact increase in CAV penetration rates on highways is currently subject to speculation and will likely be determined by future technological, societal, and market developments, such as price decline, customer willingness to pay for automated driving features, or the extent to which shared mobility concepts can replace private car ownership (Bansal & Kockelman, 2017; Litman, 2021; Noy et al., 2018). It is therefore reasonable to assume that there will be mixed traffic on the highway in the foreseeable future, in which the proportion of manual drivers will (slowly) decrease. Thus, manual drivers can be expected to be in the majority in mixed traffic in the coming decades (e.g., Litman, 2021; Milakis et al., 2017; Stange, 2021; Zmud et al., 2019).

For manual drivers on highways in Germany, driving in mixed traffic will be a novel experience. Automated vehicles are expected to have a different driving strategy than humans, especially in terms of rule-compliance (Fuest et al., 2020; Nyholm & Smids, 2020; Stange et al., 2022), and automated vehicles will presumably lack the type social understanding that guides human behavior in traffic (Hancock, 2020). Therefore, manual drivers' expectations of other drivers' driving behavior gained in non-automated traffic do not necessarily apply to automated vehicles (van Loon & Martens, 2015). This may lead to safety-critical interactions in mixed traffic, as manual drivers have incorrect expectations of automated driving behavior and may therefore be surprised by their rule-compliant driving behavior (Stange et al., 2022; Stanton et al., 2020).

Despite these challenges and the fact that car manufacturers intend to introduce the first generation of Level 3 on highways in Germany in the near future (see e.g., Audi, 2017; Daimler, 2019, 2021), the perspective of manual drivers has been insufficiently addressed in previous human factors research. The available studies that have addressed the perspective of manual drivers in mixed traffic have focused on the distinguishability of automated and human-driven vehicles (HDVs) based on driving behavior (Stange et al., 2022; Stanton et al., 2020), as well as the effects of automated driving behavior (Mahdinia et al., 2021; Stange et al., 2022), platooning (Gouy et al., 2013, 2014), external marking (Fuest et al., 2020; Stange et al., 2022), and dedicated lanes for automated vehicles (Razmi Rad et al., 2021; Schoenmakers et al., 2021) on the experience and driving behavior of manual drivers in mixed traffic.

In a driving simulator study, Stange et al. (2022) investigated manual drivers' first contact with CAVs in four typical driving situations on the highway, including 1) entering the highway, 2) overtaking, 3) letting a vehicle merge, and 4) reacting to the introduction of speed limits. The study showed that manual drivers were able to distinguish CAVs from HDVs based on the difference in driving behavior. Moreover, manual drivers rated interactions with CAVs as safe and as comfortable as interactions with HDVs. At the same time, however, manual drivers were surprised by the CAV's strict adherence to the speed limits, which in some cases resulted in short time headways of the following manual driver. However, an external marking of the CAV's current automated driving mode (status HMI) had no effect on manual drivers' driving behavior or subjective experience in mixed traffic. A driving simulator study by Fuest et al. (2020) yielded similar results. In the three highway driving scenarios (roadworks, traffic jam, lane change) studied, the external marking of the CAVs (status eHMI) had no effect on manual drivers' subjective ratings or driving behavior. In an online study with videos from real-world road traffic, Stanton et al. (2020) found different results regarding the distinguishability of driving modes from a manual driver's external perspective. The authors investigated whether manual drivers could identify the driving mode of an unmarked Tesla Model S (automated vs. human-driven) based on its lane-changing behavior on the highway. The results show that the two driving modes were "virtually indistinguishable" (p. 422) from a manual driver's external perspective. The authors therefore suggested the use of an external marking to communicate the vehicle's current driving mode (status eHMI) to surrounding manual drivers in mixed traffic, so that manual drivers can adjust their expectations and driving behavior to this different type of vehicle.

However, the studies described focused on brief interactions between manual drivers and CAVs in a few selected driving situations. Considering that drivers usually travel longer distances on the highway rather than experiencing single driving situations, the investigation of repeated interactions with CAVs in mixed traffic over longer distances seems essential. In these repeated interactions, it is yet unclear to which extent the presence of (multiple) CAVs affects the subjective experience and driving behavior of manual drivers in mixed traffic.

At the perceptual level, drivers can be expected to successively build a mental model of how CAVs typically behave in frequently occurring space-sharing conflicts during repeated interactions with CAVs over longer highway segments and over time (Endsley, 1995; Markkula et al., 2020). With sufficient experience in mixed traffic, manual drivers can then use their mental model, including the schemata of CAV driving behavior, as part of their situational awareness to correctly and reliably anticipate the future behavior of CAVs in space-sharing conflicts (see Baumann & Krems, 2009; Endsley, 1995). Situation awareness is defined as "[1] the perception of elements in the environment within a volume of time and space, [2] the comprehension of their meaning, and [3] the projection of their status in the near future." (Endsley, 1988, p. 97). Thus, the manual driver's anticipation of the next maneuver of a CAV is the highest level of situational awareness. Eventually, manual drivers might be able to mitigate safety-critical interactions in mixed traffic when following a CAV by, for example, maintaining a larger distance from the lead CAV at a speed limit, knowing that CAVs will brake early (see Stange et al., 2022).

As CAVs become more prevalent in mixed traffic, it may even be possible for the manual driver to not only anticipate the driving behavior of the CAV, but also begin to adapt their own driving behavior to that of the CAV. In a driving simulator study, Razmi Rad

Table 1

Speed limits (80 km/h, 100 km/h, 130 km/h) and distances (0 m – 35000 m) on the ten subsections (1 – 10) per highway section (0 %, 25%, 50%, 75% CAVs in mixed traffic).

Penetration rates of HAVs on each highway section (within-subjects)					
			0%	25%	50%
Subsection	Distance	Speed limit			75%
1	0 m – 5000 m	130 km/h			
2	5000 m – 5500 m	80 km/h			
3	5500 m – 10000 m	130 km/h			
4	10000 m – 10500 m	100 km/h			
5	10500 m – 20000 m	130 km/h			
6	20000 m – 20500 m	80 km/h			Same as “0%”
7	20500 m – 30000 m	130 km/h			
8	30000 m – 30500 m	100 km/h			
9	30500 m – 34500 m	130 km/h			
10	34500 m – 35000 m	80 km/h			

et al., (2021) investigated the effect of a dedicated lane for CAVs on the driving behavior of manual drivers, assuming a penetration rate of 43% CAVs in mixed traffic. Results show that the dedicated CAV lane affected driving behavior of manual driver on the neighbour lane, as manual drivers' imitated the CAVs' platooning behavior (shorter time headways to lead vehicles, shorter accepted merging gaps). The authors explain these findings with the high conspicuity of the CAVs on the dedicated lane. Schoenmakers et al. (2021) found similar results, with manual drivers maintaining shorter time headways to vehicles ahead when driving near a CAV platoon. In addition, male drivers maintained shorter time headways than female drivers in the driving simulator study (Schoenmakers et al., 2021). In a field experiment, Mahdinia et al. (2021) examined the driving behavior of manual drivers following a lead vehicle with an automated or human speed profile. Results showed reductions in manual drivers' speed and acceleration volatility as well as larger time-to-collisions, i.e. safer distances, when following the vehicle with an automated speed profile compared to the vehicle with a human driver speed profile.

Thus, the (prominent) presence of CAVs in mixed traffic can affect manual drivers' behavior in terms of distance keeping and speed behavior. However, it remains to be investigated whether this role model effect of CAVs on manual drivers can be transferred to other aspects of driving behavior, such as compliance with speed limits or even rule-following in general.

Using a status eHMI enables manual drivers in mixed traffic to identify Level 3 vehicles immediately and to activate schemata with stored information about CAVs, such as their typical driving behavior (Schieben et al., 2019). Based on these activated schemata, manual drivers can adjust their expectations and react appropriately (Schieben et al., 2019; Stanton et al., 2020). Thus, marking vehicles in automated driving mode via a status eHMI may promote the adaptation of safe driving behavior and the development of an appropriate mental model of CAV driving behavior in manual drivers. For example, manual drivers learn that CAVs adhere too strictly to the speed limits, so manual drivers following the CAV might increase the safety margin to a lead CAV before the introduction of a speed limit because manual drivers know that the CAV will decelerate early to reach the maximum permitted speed at the road sign. In this way, a status eHMI might mitigate such “surprise effects” for following manual drivers caused by the CAVs' rule-compliance (see Stange et al., 2022).

However, this potential benefit of a status eHMI in the establishment of an adequate mental model of CAVs as part of manual drivers' situation awareness might first unfold in repeated interactions, not at first contact, because these schemata for CAV may have to be learned over time. It is therefore not surprising that the driving simulator studies described previously found no positive effects of status eHMIs on the experience or driving behavior of manual drivers at first contact with CAVs in mixed traffic (Fuest et al., 2020; Stange et al., 2022). Therefore, we address the use of a status eHMIs in the present research.

Based on these considerations, the present study examined how manual drivers experience and react in repeated interactions with multiple CAVs on longer highway sections. To this end, we conducted a driving simulator study in which manual drivers repeatedly interacted with Level 3 vehicles in mixed traffic on four highway sections. On these sections, we varied the penetration rate CAVs (within-subjects factor), and we examined the effects of a status eHMI compared to no eHMI (between-subjects factor). We addressed the following research questions:

- 1 Do manual drivers notice differences in CAV penetration rates in mixed traffic on the highway sections from their external perspective?
- 2 Do increasing CAV penetration rates in mixed traffic affect manual drivers' perceived safety, and comfort?
- 3 Are CAVs perceived by manual drivers as an obstacle or as a role model in terms of rule-compliant and defensive driving behavior?
- 4 Do manual drivers adapt their driving behavior in response to an increasing CAV penetration rate?
- 5 Does a status eHMI affect manual drivers' perceived safety and comfort or their driving behavior in mixed traffic?

Due to the explorative approach of this study, we present no directed hypotheses for these research questions.

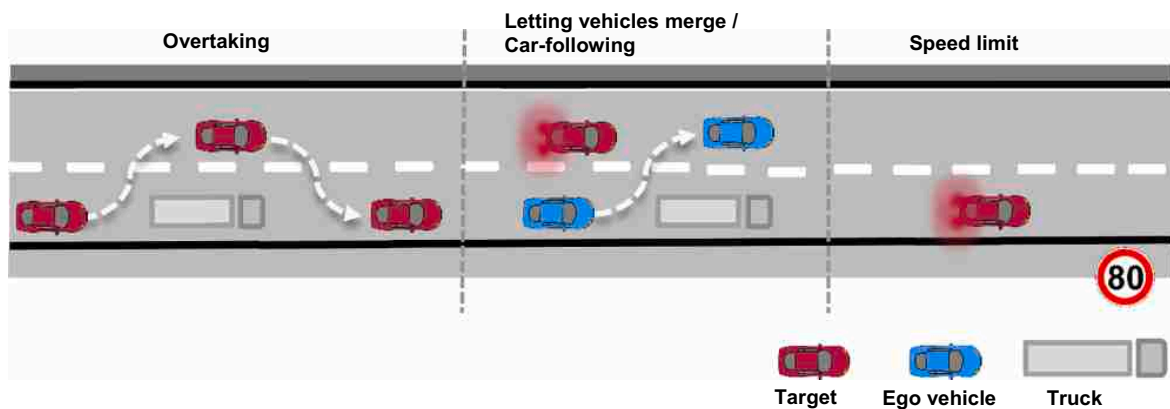


Fig. 1. Schematic overview of driving situations on highway sections, including overtaking slower vehicles, letting vehicles merge / car-following, and introduction of speed limits, with the participant in the ego vehicle (blue vehicle) interacting with the target (red vehicle) or a truck (gray vehicle).



Fig. 2. Fixed-base driving simulator of the Department of Traffic and Engineering Psychology at the TU Braunschweig (Photo: Frauke L. Berghöfer, with permission, [Stange et al., 2022](#)).

2. Methods and materials

In preparation of the present study, we interviewed $N = 9$ experts from the automotive industry, i.e., OEMs ($n = 4$), and suppliers operating in Germany ($n = 4$), as well as from academia ($n = 1$) to implement realistic highway sections (see section 2.1) with plausible configurations of conditionally automated and human driving behavior (see section 2.2) and external marking of CAVs (see section 2.3; see also [Stange et al., 2022](#)) in the driving simulator.

2.1. Creating the highway sections in the driving simulator

We used a typical German highway with two lanes per direction of travel as the basis for the highway sections. Each highway section had a length of 35 km and was completed by the participants in approximately 20 to 25 min. Each highway section consisted of ten subsections with three different speed limits (80 km/h, 100 km/h, 130 km/h; see [Table 1](#)), which are typical on highways in Germany. The speed limits were introduced by speed signs and shown to the participants via a head-up display, so that the participants were informed about the current speed limit at all times. The highway sections were identical in terms of road design. In the sections where 130 km/h was allowed, the speed of the CAV varied between 100 km/h and 110 km/h, as the experts interviewed assumed that these CAVs would drive rather defensively and not as fast as allowed. Each drive started at a highway rest area, then participants completed the drive on the highway section and ended at another highway rest area. The speed of the ego vehicle was limited to 150 km/h to minimize the risk of motion sickness. The order of the highway sections was not randomized between participants, instead all participants completed the same order.

The highway sections included only driving situations that the interviewed experts believed first-generation Level 3 vehicles could handle without sending a take-over request to the passenger, including overtaking slower vehicles, car-following, and speed adjustments in response to the introduction of speed limits (see [Fig. 1](#); see also [Stange et al., 2022](#)), once their functionality extends beyond

Table 2

Overview of driving parameters (speed in km/h, minimum safety distance from vehicles ahead in time headway) used to parameterize CAV and HDV driving behavior in the driving simulation.

Parameter	Vehicle type	
	CAV	HDV
Speed behavior:	Target speed:	Target speed:
130 km/h zone	110 km/h – 110 km/h	130 km/h
100 km/h zone	100 km/h	130 km/h
80 km/h zone	80 km/h	130 km/h
Safety distance from vehicles ahead	Min. 2.75 s time-headway (all subsections)	Min. 1.2 s time headway (all subsections)

driving in congestion. In principle, these driving scenarios also apply to vehicles in Level 2 driving mode (partial automation; [SAE International, 2021](#)). We excluded all driving situations in which, according to the experts we interviewed, the system limits of a first-generation CAV could be reached, including construction sites, highway intersections, highway on- and off-ramps, missing lane markings, and extreme weather conditions.

The driving simulation was programmed using SILAB 6.0 (WIVW GmbH, <https://www.wivw.de/>, see also [Krueger et al., 2005](#)) and the study was conducted in the fixed-based driving simulator of the Department of Traffic and Engineering Psychology at the TU Braunschweig. The driving simulator consists of a physical mock-up seat box with driver and passenger seats (see [Fig. 2](#)). The three LCD projectors (1920 × 1080 pixels each) projected the scenery onto an array of three screens (2 m × 2 m each) covering a field of view of 180°. The rearview mirror was projected onto the front screen. Separate screens (seven inches each) simulated the speedometer the side mirrors. The experimenter monitored the simulator drives on six computer screens in an adjacent room. Driving data, including the ego vehicle's current speed, position, and distances to surrounding simulated vehicles, were recorded at a rate of 60 Hz.

2.2. Defining conditionally automated and human driving behavior on the highway sections

The experts agreed that the CAVs will strictly comply with the speed limits. On the highway sections with speed limits of 80 km/h and 100 km/h, the simulated CAVs therefore adhered to the specified maximum speed. On the sections with speed limit of 130 km/h, the CAVs' target speed was set in the range between 100 km/h and 110 km/h (relaxed and defensive driving style, see section 2.1). This was also done to allow participants to approach and overtake these vehicles, as driving at higher speeds in the driving simulator may cause or intensify motion sickness. In sections where the speed limit was reduced to 80 km/h or to 100 km/h, the CAVs reached the maximum permitted speed at the speed sign and accelerated only after passing the corresponding road sign at the end of this speed limit section. The experts also agreed that the size of the safety margins that CAVs will maintain from lead vehicles is likely to be larger than the safety margin of half a speedometer recommended in Germany (>1.8 s time headway; see [Vogel, 2003](#)). Therefore, the simulated CAVs' time headways to lead vehicles was set to a target minimum of 2.75 s. Moreover, the experts stated that CAVs will drive in the center of the lane with minimal lateral deviations only, so we implemented a perfect lane centering for the simulated CAVs. In principle, these driving behavior parameter may also apply to vehicles in Level 2 driving mode (partial automation; [SAE International, 2021](#)).

Regarding the driving behavior of the HDVs, we implemented a predefined target speed of 130 km/h on all subsections. The HDVs' predefined safety margin to the lead vehicle was at least 1.2 s time headway. Therefore, when speed limits were introduced, the simulated HDVs could exceed the maximum permitted speed if there was no lead vehicle in front of them. In terms of lane centering, the simulated HDVs deviated up to one meter to the right and left from the center of the lane. The driving parameters of the CAVs and HDVs are presented in comparison in [Table 2](#).

Regarding the configuration of the speed behavior of CAVs and HDVs, it should be noted that the values reported in [Table 2](#) and described above were target speeds, reached only when the minimum safety distance to the vehicle ahead could be maintained. Once the safety distance fell below the minimum distance to the vehicle ahead, the CAV or HDV adjusted the speed to restore and maintain the distance. As a result, not all HDVs drove the target speed of 130 km/h during the entire simulator drive, but adapted their driving behavior to the behavior of the vehicles ahead and to the behavior of the participant in the ego vehicle. In this way, setting a target speed introduced some variation in the speed behavior of HDVs. Despite the different configurations of the vehicles, which were modeled after automated and human behavior, all simulated vehicles were controlled by an algorithm with predefined values. Therefore, the distinction between automated and human-driven vehicles in this study represents a distinction between strictly rule-based and less strict driving styles.

To create a natural traffic flow and to ensure that interactions between the participant in the ego vehicle and surrounding vehicles occurred, we used traffic sources to place multiple simulated CAVs and HDVs in front of and behind the ego vehicle in the driving simulation, aiming for a medium traffic density. We also placed eight trucks driving at 80 km/h in the right lane to encourage participants in the ego vehicle to overtake these trucks and change lanes.

2.3. Marking the CAVs on the highway sections

The idea of an external marking for CAVs is currently being discussed in human factors research (see e.g., [Dey et al., 2020](#); [Schieben](#)



Fig. 3. A Level 3 vehicle on the highway, marked externally by a blue light rectangle (status eHMI; Photo: Doris Sonntag, with permission; Stange, 2021).

Table 3

Experimental design including the between-subjects factor external marking and the within-subjects factor penetration rate as well as the number of participants in each experimental condition.

External marking (between-subjects factor)	Penetration rate (within-subjects factor)			
	0%	25%	50%	75%
Status eHMI ($n = 17$)	8 male, 9 female, $M = 38.2$ years ($SD = 17.4$)			
No eHMI ($n = 17$)	10 male, 7 female, $M = 44.0$ years ($SD = 17.9$)			
Control group ($n = 17$)	11 male, 6 female, $M = 41.5$ years ($SD = 18.2$)			

et al., 2019; Stanton et al., 2020). Therefore, the experts we interviewed suggested including an eHMI in the driving simulator study. Based on our results from a previous driving simulator study that investigated manual drivers' first contact with CAVs in mixed traffic on the highway (see Stange et al., 2022), the status eHMI in the present study referred to the vehicle's current automated driving mode (status eHMI). We used a blue light rectangle, which was visible in a 360° radius around the vehicle (see Fig. 3; see Stanton et al., 2020).

2.4. Independent variables in the study

The study followed a 3×4 mixed design (see Table 3). We implemented four highway sections (see Table 1) in the driving simulator, each with a different CAV penetration rate (within-subjects factor), from 0% to 75% in steps of 25%. This variation in CAV penetration rate applied only to vehicles traveling into the same direction as the ego vehicle, while the opposite lanes were filled with random traffic to create a natural driving experience for participants.

Participants were randomly assigned into one of three experimental groups that differed in the external marking of the CAVs (between-subjects factor). In the "no eHMI" group, neither the CAVs nor the HDVs were marked. We instructed participants that CAVs and HDVs were indistinguishable from the external perspective. In the "status eHMI" group, CAVs were marked, while the HDVs were unmarked. This allowed participants to use the eHMI as a visual cue to distinguish the CAVs from the HDVs. Unlike the previous two groups, participants in the "control group" were not instructed about the mixed traffic on the highway sections because the control group served as a manipulation check to see if the differences in the driving behavior of the simulated vehicles were noticeable to the participants. Therefore, we instructed participants in this group that we wanted to examine traffic flow and safety in manual traffic as a cover story. After completing the highway drives, we conducted a debriefing with the participants in the control group.

2.5. Instructions to the participants

Before driving in the driving simulator, participants in the "status eHMI" and the "no eHMI" groups received detailed information about typical CAV driving behaviors, including speed behavior and distance keeping, which were relevant to the driving scenarios investigated in the present study (see Fig. 1). To this end, participants in these groups watched three pre-recorded videos from a previous driving simulator study (see Stange et al., 2022), in which we used the same driving situations as in the present study. In the first video, participants were shown a CAV strictly obeying the speed limits by slowing down in front of the speed sign to reduce its speed to the maximum speed allowed at the sign. In the second video, participants saw a CAV maintaining a safe distance of about half a speedometer from other vehicles ahead. An HDV then reduces this distance by cutting in right in front of the CAV. The CAV then reduces its speed to restore the minimum safe distance to the HDV ahead. In the third video, participants saw a CAV independently overtaking slower vehicles.

We introduced the information phase because we assume that manual drivers will have at least some (theoretical) knowledge about CAV driving behavior once the penetration rate of these vehicles in mixed traffic is 25% or higher. In this way, we created some shared knowledge about CAV driving behavior and encouraged the building of appropriate mental models of CAV driving behavior. The information phase also served to familiarize participants in the “status eHMI” group with the eHMI design used in the study, as CAVs were marked in the “status eHMI” group videos.

In addition, all participants were instructed to drive on the highway sections as they normally would if they were driving their own vehicle on the highway. This included speeding and driving below the recommended minimum safety margin to lead vehicles. Participants were also reminded of the right-hand driving rule which applies on highways in Germany and to avoid collisions with surrounding vehicles, just as they would in real-world road traffic.

2.6. Dependent variables in the study

Since we aimed to examine manual drivers' experience and reactions to driving in mixed traffic, we collected questionnaire data and driving data in this study. In addition, the entire simulator drives were recorded using a video camera (GoPro Hero 3+) mounted on the simulator dashboard and set to wide angle to film the entire scenery from the perspective of the participant in the ego vehicle. This was done to facilitate the detection of errors in the driving data recording and in the manipulation of the driving behavior of the simulated vehicles.

We measured participants' perceived penetration rate of CAVs on each highway section (“What percentage of the vehicles traveling in the same direction as you were automated vehicles?”) on a scale from 0% to 100%. This question was asked only in the “no eHMI” and the “status eHMI” groups, where participants were informed about mixed traffic. We measured perceived safety (“What category best describes your perception of the traffic situation on the previous highway section?”) on an 8-point scale from 1 (“harmless”) to 8 (“situation not acceptable”) after each highway section. The scale is based on the controllability scale by Neukum et al. (2008) and was adapted from Stange et al. (2022). We measured comfort (“How comfortable was the drive on the previous highway section?”) using a 5-point Likert scale ranging from 1 (“very comfortable”) to 5 (“very uncomfortable”) after each highway section. To measure perceived efficiency (“How well did you progress on the previous highway section?”) after each highway section, we used 5-point Likert scale ranging from 1 (“very inefficient”) to 5 (“very efficient”).

To analyze manual drivers' behavioral adaptation to mixed traffic regarding speed behavior, we recorded participants' average speed on all ten subsections of each highway section and averaged these mean values for each of the three speed zones (80 km/h, 100 km/h, 130 km/h) on each highway section. We analyzed participants' safety margins (measured by mean and minimum time headway) to lead vehicles as indicators of safety criticality in mixed traffic interactions. In the analysis of safety criticality, we only considered interactions in which a lead vehicle was present (< 3 s time headway between the ego vehicle and the lead vehicle). We also excluded extremely short time headways (< 0.1 s), as these were caused by errors in driving data recording. In addition, we calculated the percentage of safety critical time headways to lead vehicles (< 1.0 s) for the ten subsections of each highway section. We then averaged the results for each of the three speed zones (80 km/h, 100 km/h, 130 km/h; see Table 1) on each highway section.

2.7. Participants

The sample included 51 participants aged 22 to 74 years ($M = 41.5$ years, $SD = 18.1$, 22 female) recruited from an internal database. Three of the initial 54 participants dropped out of the study due to motion sickness during the training drive. Participants in this study held their driving license for 24.0 years ($SD = 19.0$) on average, and 23 (45%) participants reported an annual mileage of more than 9000 km/h. The majority of participants (61%) reported driving a car at least several times a week. Forty (78%) participants had used driving assistance systems before, and 36 (71%) participants had driven in the driving simulator at least once before the present study. The study took approximately two hours to complete and participants received monetary compensation for their participation. Undergraduate Psychology student could choose between monetary compensation or course credit. The study complied with the German Psychological Societies' adaptation of the American Psychological Association's Code of Ethics and was approved by the Ethics Committee of the Faculty of Life Sciences at the TU Braunschweig. Before giving informed consent for participation and to the scientific use of their data, participants were informed about the aim of the study, which was to examine the effects of mixed traffic on the experiences and perceptions of human drivers on the highway. Participants were further informed that they were free to drop out of the study at any point without any disadvantages or loss of their monetary compensation.

2.8. Data treatment and analysis

2.8.1. Questionnaire data

We performed 3×4 mixed ANOVAs with post-hoc pairwise comparisons for the dependent variables perceived safety, comfort, and perceived efficiency. There was no missing data in the analyses. The analysis of the dependent variable perceived safety revealed that $n = 12$ ($n_{\text{status eHMI}} = 5$, $n_{\text{no HMI}} = 4$, $n_{\text{control group}} = 3$) manual drivers rated driving in non-automated traffic without CAVs as *dangerous* or *situation not acceptable* on the 8-point scale. Further analysis revealed no evidence that characteristics of the sample affected perceived safety ratings. These participants had no less driving experience (in a driving simulator) than the rest of the sample. In addition, the analysis of the video data provided no indication of any safety critical situations during the simulator drives within this group of participants. Thus, it can be assumed that these participants either found driving in the driving simulator and / or driving on the highway *dangerous* or *not acceptable* per se, or they misunderstood the safety scale using an incorrect reference system (see section 4.2

Table 4

ANOVA results for all interaction and main effects of the between-subjects factor external marking and the within-subjects factor penetration rate on perceived safety, comfort, perceived efficiency, and perceived penetration rate (significant p-values in bold).

	Effect	<i>F</i>	<i>df</i>	<i>p</i>	η_p^2
Perceived safety*	P (Penetration rate)	3.6	2,3,83.1	.026	.09
	E (External marking)	0.6	2,36	.585	
	P × E	0.4	4,6,83.1	.812	
Comfort*	P	2.9	2,7,96.9	.047	.07
	E	0.1	2,36	.948	
	P × E	1.2	5,4,96.9	.333	
Perceived efficiency*	P	4.1	2,5,91.6	.012	.10
	E	0.5	2,36	.603	
	P × E	0.8	5,1,91.6	.544	
Perceived penetration rate**	P	14.7	2,8,64.0	< .001	.39
	E	0.9	1,36	.362	
	P × E	15.4	2,8,64.0	< .001	

Note. **N* = 39. ***N* = 25.

for further discussion). Therefore, the data sets of these participants were excluded from further analysis of the questionnaire data. Thus, data from 156 simulator drives (4 highway sections × 39 participants) were included in the analysis of the questionnaire data, representing 76.5% of all questionnaire data.

We performed a 2 × 4 mixed ANOVA with pairwise comparisons for the dependent variable perceived penetration rate. As participants in control group were unaware of the mixed traffic, this variable was measured only in the “no eHMI” and the “status eHMI” groups, reducing the sample size from *N* = 51 participants to *n* = 34 participants (*n*_{status eHMI} = 17 participants, *n*_{no eHMI} = 17 participants). As we excluded the data sets of participants rating the simulator drive in non-automated traffic as *dangerous* or *not acceptable* from the analysis of questionnaire data, we excluded data of *n* = 9 participants from the analysis of perceived penetration rate. Thus, we analyzed questionnaire data from 88 simulator drives completed by *n* = 25 participants (*n*_{status eHMI} = 12 participants, *n*_{no eHMI} = 13 participants), representing to 73.5% of all questionnaire data. In the “status eHMI” group, the perceived penetration rate was not measured in the 0% condition because participants in this group were aware that there were no CAVs present on this highway section. In total, questionnaire data were collected from 119 simulator drives completed by *n* = 34 participants.

2.8.2. Driving data

Driving data were recorded from 204 simulator drives (4 highway sections × 51 participants). The data sets of *n* = 2 participants were missing due to technical issues during data recording. In total, driving data from 196 simulator drives completed by *n* = 49 participants were analyzed, representing 96% of all driving data. The last subsection (34500 m – 35000 m) of each highway section was excluded from the driving data analysis because most participants decelerated in this section to reach the highway rest area.

We performed 3 × 4 mixed ANOVAs with post-hoc pairwise comparisons for the dependent variables average speed and the safety criticality measures, separately for each of the three speed zones (80 km/h, 100 km/h, 130 km/h). Thus, we performed three analyses per dependent variable. The safety criticality measures mean and minimum time headway were analyzed only if a lead vehicle was present (see section 2.6). Therefore, mean and minimum time headways were calculated for *n* = 49 participants in the 130 km/h speed zone, *n* = 27 participants in the 100 km/h speed zone, and *n* = 30 participants in the 80 km/h speed zone.

IBM SPSS Statistics 25 was used for statistical data analysis. All mean values reported are with 95% confidence intervals (CI). A test level of $\alpha = 0.05$ was applied to all statistical tests. For significant results of ANOVAs, η_p^2 is given as the effect size. When the assumption of sphericity was violated in ANOVAs, degrees of freedom were corrected with either a Greenhouse-Geisser correction ($\epsilon < .75$) or a Huynh-Feldt correction ($\epsilon > .75$). In regard of the exploratory approach of the study, the test level was not adjusted in order to better detect relevant effects, while at the same time minimizing the interpretation of random variation.

3. Results

3.1. Manipulation check

As a manipulation check of the within-subjects factor penetration rate, we recorded the actual number of CAVs and HDVs on each highway section in the driving simulation and compared these numbers to intended number of CAVs and HDVs in the experimental design. In the 25% automation condition, 23.4% of the simulated vehicles were CAVs. In the 50% and 75% automation conditions, the deviations from experimental design were somewhat larger. In the 50% automation condition, the actual penetration rate of CAVs was 44.1%, and in the 75 % automation condition, an average of 61.0% of the vehicles encountered by the manual drivers on the highway sections were CAVs. Thus, the manipulation of the within-subjects factor penetration rate was successful for the 25% and 50% conditions and partly successful for the 75% condition. The actual number of simulated vehicles rendered by the simulation differs from the number of simulated vehicles provided in the experimental design because we used vehicle sources instead of rendering each simulated vehicle individually. In addition, we only counted the simulated vehicles that participants could see in the simulated environment, either through the windshield, windows, or rearview mirror. As a result, the number of vehicles seen varied between participants, depending on the individual driving behavior, especially their speed and overtaking behavior.

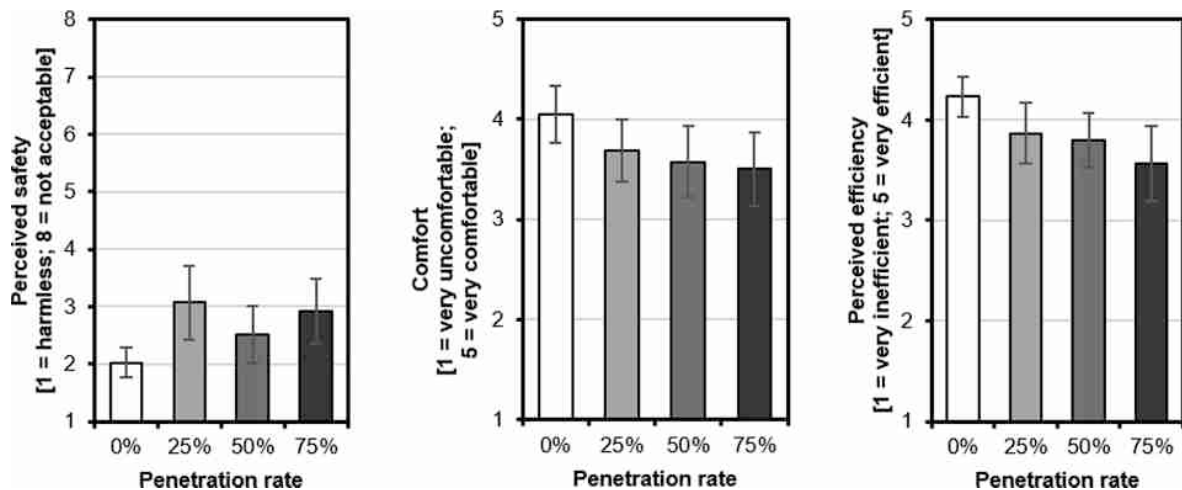


Fig. 4. Subjective ratings of perceived safety (left), comfort (middle), and perceived efficiency (right; all means with 95% CI) depending on CAV penetration rate (0%, 25%, 50%, 75%) on the four highway sections.

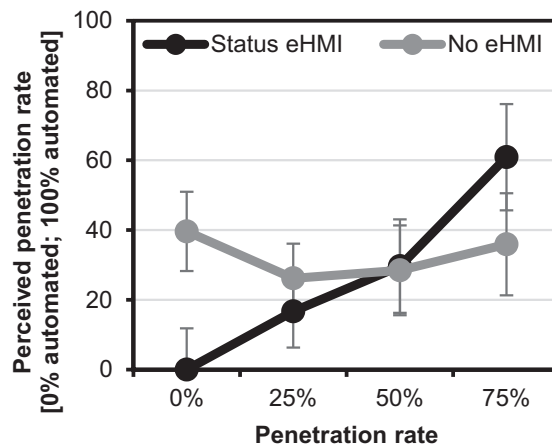


Fig. 5. Perceived penetration rate ratings (means with 95% CI) depending on CAV penetration rate (0%, 25%, 50%, 75%) and external marking (status eHMI, no eHMI).

3.2. Subjective ratings

After each highway section, participants completed a questionnaire assessing their subjective experience of the previous drive in mixed traffic. The results of the statistical tests performed for the dependent variables perceived safety, comfort, perceived efficiency, and perceived penetration rate are reported in Table 4. The analyses revealed significant main effects of the within-subjects factor penetration rate on all four dependent variables. In addition, there was a significant interaction effect between penetration rate and external marking on the ratings of perceived penetration rate.

Fig. 4 illustrates the significant main effects of CAV penetration rate on participants' ratings of perceived safety (left), comfort (middle), and perceived efficiency (right). Perceived safety ratings were lowest (= safest) in the 0% automation condition, being rated *a little uncomfortable* ($M = 2.0$). This was followed by the 50% ($M = 2.5$) and 75% ($M = 2.9$) automation conditions. The 25% automation condition was rated as the most uncomfortable ($M = 3.1$). Pairwise comparisons revealed significant differences between the 0% automation condition and the conditions with 25% and 75% CAVs, as well as a trend for the 50% automation condition ($p = .077$; see Table A1 in the Appendix for all pairwise comparisons). Ratings of perceived safety changed by approximately one scale point on the 8-point perceived safety scale, as penetration rates increased on the highway sections, with all mean ratings being located in the lower half of the scale and ranging between the labels *a little uncomfortable* ($=2$) and *medium uncomfortable* ($=3$).

Comfort ratings were highest in the 0% automation condition ($M = 4.1$), being rated *rather comfortable*. This was followed by the 25% and 50% automation conditions ($M = 3.7$, $M = 3.6$). The 75% condition was rated as the least comfortable ($M = 3.5$). Thus, manual drivers rated driving in non-automated traffic significantly more comfortable than driving in mixed traffic, regardless of the exact penetration rate of CAVs. Pairwise comparisons revealed significant differences between the non-automated traffic condition and

Table 5

ANOVA results for all interaction and main effects of the between-subjects factor external marking and the within-subjects factor penetration rate on the average speed (130 km/h, 100 km/h, 80 km/h; significant p-values in bold).

	Effect	<i>F</i>	<i>df</i>	<i>p</i>	η_p^2
130 km/h	P (Penetration rate)	4.8	1,777.3	.015	.10
	E (External marking)	0.3	2,46	.754	
	P × E	0.6	3,477.3	.654	
100 km/h	P	4.2	2,3105.8	.014	.08
	E	1.0	2,46	.369	
	P × E	1.6	4,6105.8	.161	
80 km/h	P	7.7	3,138	< .001	.14
	E	1.1	2,46	.351	
	P × E	0.5	6,138	.806	

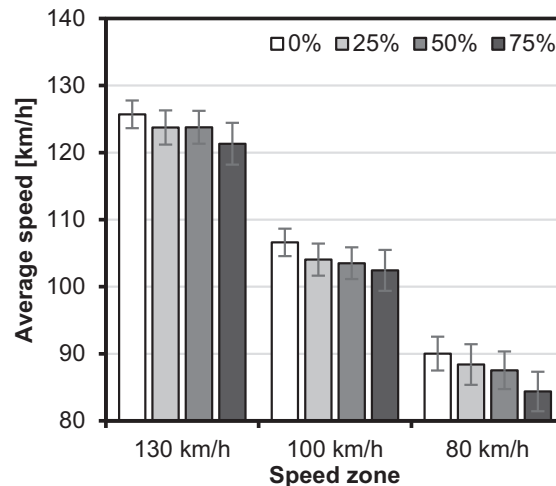


Fig. 6. Participants' average speed in 130 km/h (left), 100 km/h (middle), and 80 km/h zones (right; all means with 95% CI) depending on CAV penetration rate (0%, 25%, 50%, 75%).

all three mixed traffic conditions (for all pairwise comparison, see Table A1 in the Appendix). Overall, comfort ratings ranged between the labels *rather comfortable* and *medium comfortable*, with all mean ratings being located in the upper half of the scale.

Similarly, perceived efficiency was rated highest in the 0% automation condition, being rated as *rather efficient* ($M = 4.2$). This was followed by the 25% ($M = 3.9$) and 50% ($M = 3.8$) automation conditions. The 75% condition was rated least efficient ($M = 3.6$). Pairwise comparisons revealed significant differences between the 0% automation condition and the other three mixed traffic conditions (see Table A1 in the Appendix for all pairwise comparisons). Overall, ratings of perceived efficiency were located in the upper half of the rating scale, with mean values changing by approximately one scale point across the four automation conditions.

To assess the distinguishability of CAVs and HDVs in mixed traffic, participants estimated the penetration rate of CAVs on each highway section. As can be seen in Fig. 5, participants in the “status eHMI” group provided more correct estimates than participants in the “no eHMI” group, with mean estimates increasing alongside the actual penetration rates. Despite the correct trend of the estimates in the “status eHMI” group, participants underestimated the penetration rate in the 25% and 50% automation conditions ($M_{25\%} = 16.7\%$, $M_{50\%} = 29.7\%$). In the 75% automation condition, participants estimated a penetration rate of $M_{75\%} = 60.9\%$ CAVs, whereas the manipulation check revealed an actual CAV penetration rate of 60.4% in this condition (see section 3). Thus, the estimate for this highway section was nearly accurate. In contrast, the mean estimates of participants in the “no eHMI” group ranged between 26% and 40% in the four automation conditions ($M_{0\%} = 39.6\%$, $M_{25\%} = 26.2\%$, $M_{50\%} = 28.5\%$, $M_{75\%} = 35.9\%$). These mean estimates were inaccurate, except for the 25% automation condition, which could be a random finding because there was no systematic pattern of estimates in the “no eHMI” group.

3.3. Driving data

3.3.1. Average speed

The results of the statistical tests performed for the dependent variable average speed, separately for each speed zone, are presented in Table 5. The analyses revealed significant main effects of the within-subjects factor penetration rate on participants' average speed in all three speed zones. Fig. 6 shows the significant main effects of the penetration rate of CAVs on the average speed in the 130 km/h, 100 km/h, and 80 km/h zones.

Table 6

ANOVA results for all interaction and main effects of the between-subjects factor external marking and the within-subjects factor penetration rate on the mean time headway, minimum time headway, and the percentage of safety critical interactions (130 km/h, 100 km/h, 80 km/h; significant p-values in bold).

		Effect	<i>F</i>	<i>df</i>	<i>p</i>	η_p^2
130 km/h	Minimum time headway*	Penetration rate (P)	1.1	2,7,124.0	.372	
		External marking (E)	0.1	2,46	.982	
		P × E	0.9	5,4,124.0	.491	
	Mean time headway*	P	7.2	3,138	< .001	.14
		E	0.5	2,46	.621	
		P × E	1.2	6,138	.298	
	Percentage of safety critical interactions*	P	3.6	1,8,81.9	.038	.08
		E	0.6	2,46	.552	
		P × E	0.3	3,6,81.9	.844	
100 km/h	Minimum time headway***	P	1.6	3,72	.194	
		E	0.4	2,24	.669	
		P × E	1.5	6,72	.182	
	Mean time headway***	P	1.6	3,72	.750	
		E	0.4	2,24	.329	
		P × E	1.5	6,72	.280	
	Percentage of safety critical interactions*	P	3.0	3,138	.034	.06
		E	1.6	2,46	.210	
		P × E	0.4	6,138	.904	
80 km/h	Minimum time headway**	P	5.5	2,7,71.6	.003	.17
		E	2.1	2,27	.148	
		P × E	0.2	5,3,71.6	.978	
	Mean time headway**	P	6.1	2,7,73.8	.001	.19
		E	2.7	2,27	.086	
		P × E	0.5	5,5,73.8	.829	
	Percentage of safety critical interactions*	P	2.1	2,3,106.2	.120	
		E	2.3	2,46	.115	
		P × E	1.5	4,6,106.2	.189	

Note. Significant p-values in bold. **N* = 49. ** *N* = 30. *** *N* = 27.

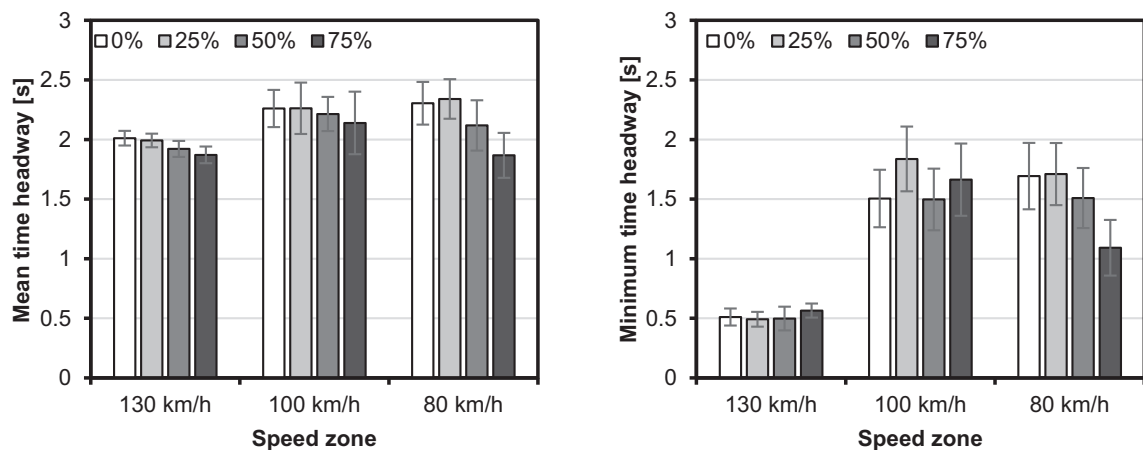


Fig. 7. Mean (left) and minimum time headway (right) in the 130 km/h, 100 km/h, and 80 km/h zones (all means with 95% CI) depending on CAV penetration rate (0%, 25%, 50%, 75%).

In the 130 km/h zone, pairwise comparisons showed that participants' average speed in the 0% automation condition ($M_{0\%} = 125.7$ km/h) was significantly higher than in the other three mixed traffic conditions ($M_{25\%} = 123.8$ km/h, $M_{50\%} = 123.8$ km/h, $M_{75\%} = 121.3$ km/h), regardless of the exact CAV penetration rate (see Table A2 in the Appendix). In addition, there was a significant difference between the 50% and 75% automation conditions ($p = .045$). As expected, the average speed of the manual drivers on all four highway sections was below the maximum permitted speed, since the predefined target speed of the simulated vehicles was max. 110 km/h (CAVs) or 130 km/h (HDVs).

A similar pattern of results also emerged in the 100 km/h zone, where participants' average speed in non-automated traffic ($M_{0\%} = 106.6$ km/h) was significantly higher than in the other three mixed traffic conditions ($M_{25\%} = 104.0$ km/h, $M_{50\%} = 103.5$ km/h, $M_{75\%} = 102.4$ km/h; see Table A2 in the Appendix). Although participants' average speed in the 75% condition changed by approximately 4 km/h compared to the non-automated traffic, the manual drivers still exceeded the speed limit of 100 km/h.

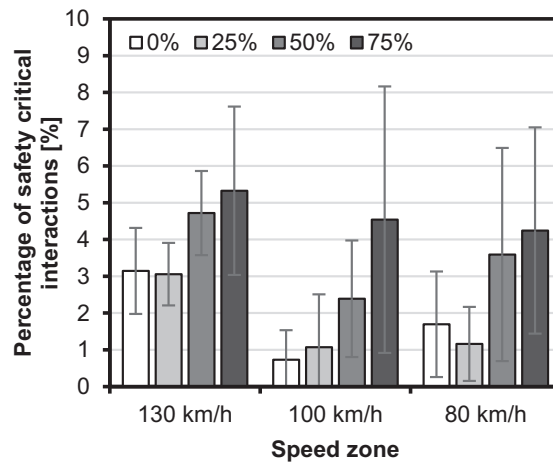


Fig. 8. Percentage of safety critical interactions in the 130 km/h, 100 km/h, and 80 km/h speed zones (all means with 95% CI) depending on CAV penetration rate (0%, 25%, 50%, 75%).

In the 80 km/h zone, pairwise comparisons showed that average speed was significantly lower in the 75% condition ($M_{75\%} = 84.4$ km/h) than in the other three conditions ($M_{0\%} = 90.0$ km/h, $M_{25\%} = 88.4$ km/h, $M_{50\%} = 87.5$ km/h; see Table A2 in the Appendix), and between 0% and 50% automation conditions. Regarding rule-compliance, manual drivers exceeded the speed limit on all four highway sections, although the average speed changed by approximately 5 km/h in the 75% automation condition compared to the non-automated traffic.

3.3.2. Safety criticality

For each speed zone, we analyzed participants' mean and minimum time headway (in seconds) to lead vehicles as a measure of safety criticality. The results of the statistical analyses are presented in Table 6. The analyses revealed significant main effects of the within-subjects factor penetration rate on the mean time headway to lead vehicles in the 130 km/h and 80 km/h zones and on the minimum time headway in the 80 km/h zone. Regarding the percentage of safety critical interactions with lead vehicles, the analysis revealed a significant main effect of penetration rate on the percentage of safety critical interactions in the 100 km/h and 130 km/h zones.

Fig. 7, on the left, depicts participants' mean time headways to lead vehicles in the three speed zones (130 km/h, 100 km/h, 80 km/h).

In the 130 km/h zone, pairwise comparisons showed that mean times headways in the 0% automation condition ($M_{0\%} = 2.0$ s) were significantly larger than in the 50% and 75% automation conditions ($M_{50\%} = 1.9$ s, $M_{75\%} = 1.9$ s; see Table A2 in the Appendix) and in the 25% automation condition ($M_{25\%} = 2.0$ s) compared to the 50% automation condition. A similar pattern emerged in the 80 km/h zone. Pairwise comparisons revealed significant differences between the 25% ($M_{25\%} = 2.3$ s) and the 50% automation conditions ($M_{50\%} = 2.1$ s, $p = .041$) as well as between the 75% automation condition ($M_{75\%} = 1.9$ s) and the other three conditions (see Table A2 in the Appendix). In the 100 km/h zone, CAV penetration rate had no significant effect on the mean time headways ($M = 2.3$ s – 2.1 s). In the three speed zones, the direction of effect was similar, with mean time headways being lowest in the 75% automation condition each. Nevertheless, all obtained mean values were above recommended minimum time headway of half a speedometer on German highways (see Vogel, 2003).

Fig. 7, on the right, shows participants' minimum time headways to lead vehicles in the three speed zones (130 km/h, 100 km/h, 80 km/h). Overall, minimum time headways in the 130 km/h zone were lower than in the other two speed zones, with minimum time headways being below the threshold for close following (< 0.9 s time headway; Vogel, 2003). In addition, the minimum time headways did not differ significantly between the four highway sections in the 130 km/h ($M = 0.5$ s – 0.6 s) and 100 km/h ($M = 1.5$ s – 1.8 s) zones. In the 80 km/h zone, pairwise comparisons revealed significant differences between the 75% automation condition ($M = 1.1$ s) and the other three conditions ($M_{0\%} = 1.7$ s, $M_{25\%} = 1.7$ s, $M_{50\%} = 1.5$ s; see Table A2 in the Appendix).

As illustrated in Fig. 8, the percentage of safety-critical interactions with lead vehicles increased along with the increasing CAV penetration rate in mixed traffic. At the same time, the mean values were subject to substantial variance. In the 130 km/h zone (see Fig. 8, on the left), pairwise comparisons showed significant differences between all conditions (see Table A2 in the Appendix), except for the 50% and 75% automation conditions ($M_{50\%} = 4.7\%$, $M_{75\%} = 5.3\%$, $p = .861$) and the 0% and 25% automation conditions ($M_{0\%} = 3.1\%$, $M_{25\%} = 3.1\%$, $p = .544$). In the 100 km/h zone (see Fig. 8, in the middle), pairwise comparisons revealed significant differences between the 0% ($M_{0\%} = 0.7\%$) and 75% automation conditions ($M_{75\%} = 4.5\%$, $p = .024$).

4. Discussion and conclusions

The aim of this study was to examine the experience and behavioral reactions of manual drivers in mixed traffic during repeated

interactions with CAVs on the highway. Participants completed four highway sections (35 km each) that differed in the penetration rate of CAVs in mixed traffic (0% – 75%). On the highway sections, the CAVs' current driving mode was either marked or not (status eHMI, no eHMI). In addition, there was a control group in which participants were unaware of the mixed traffic. In the following sections, we summarize the results and discuss them regarding the research questions.

4.1. Penetration rate estimates in mixed traffic

After each highway section, participants in “status eHMI” and the “no eHMI” groups estimated the CAV penetration rate on the previous highway section. In the “no eHMI” group, drivers estimated a penetration rate of 30% to 40% CAVs on all four highway sections, including the 0% automation section. This finding is consistent with a previous video study by Stanton et al. (2020), who suggested that CAVs and HDVs are indistinguishable based on their driving behavior from the external perspective of a manual driver. At the same time, this finding seems to contradict the results from a previous driving simulator study investigating manual drivers' first contact with CAVs (see Stange et al., 2022), in which manual drivers were able to distinguish the two driving modes based on behavioral differences. One explanation for these different results may lie in the methodological differences between the two studies. While Stange et al. (2022) examined manual drivers' interactions with one CAV or HDV at a time, the present study focused on repeated interactions during longer drives, including interactions with multiple CAVs and HDVs on each of the four highway sections. Therefore, drivers may have been less focused on each vehicle interaction and observed the driving behavior of multiple simulated vehicles simultaneously for only a few moments. Instead, drivers may have been more focused on progressing along the highway sections. Therefore, it is reasonable to assume that the observation time per vehicle was insufficient to detect behavioural differences between the two types of vehicles. This result highlights the importance of a status eHMI as a visual cue for estimating the penetration rate when driving on longer highway sections in mixed traffic.

4.2. Manual drivers' safety and comfort in mixed traffic

Regarding perceived safety and comfort, manual drivers rated driving in non-automated traffic significantly more comfortable and somewhat safer than driving on highway sections with mixed traffic. Despite these significant differences in the perceived safety ratings between non-automated traffic and mixed traffic, the mean ratings of the three highway sections with mixed traffic were still in an acceptable range, as all mean ratings at most equaled to the label *very uncomfortable*, ranging at the lower (positive) end of the scale of perceived safety (see Fig. 4, on the left) and *rather comfortable* on the comfort scale (see Fig. 4, in the middle). Thus, increasing CAV penetration rates in mixed traffic led to a reduction of manual drivers' perceived safety, and comfort, but only to a limited extent.

Since the external marking had no effect on manual drivers' perceptions of safety or comfort (see Table 4), the subjective ratings varied as a function of CAV penetration rate, regardless of the experimental group, to which participants were assigned. That is, the subjective ratings changed regardless of whether drivers knew CAVs were present on the highway sections, and whether or not they were able to detect CAVs based on the external markings. Therefore, the subjective ratings of perceived safety and comfort are attributable to manual drivers experiencing the defensive driving behavior of CAVs, which manual drivers rated as more uncomfortable and less safe than driving in non-automated traffic. So, an external HMI that allowed manual drivers to uniquely identify CAVs did not affect their perceptions of safety and comfort.

Since the assumptions made about CAV driving behavior based on expert interviews also apply to partially automated vehicles (Level 2; SAE International, 2021), these findings are equally applicable to mixed traffic interactions between manual drivers and Level 2 vehicles.

4.3. Do CAVs have a role model function for manual drivers?

The penetration rate of CAVs affected manual drivers' behavior in several ways. Manual drivers' average speed decreased significantly in all three speed zones, as the penetration rate of CAVs increased (see Fig. 6). In terms of rule-compliance, the speed-reducing effect led to nearly rule-compliant average speeds in the 80 km/h and 100 km/h zones in the 75% automation condition. However, in the 130 km/h, even without CAVs (0% penetration rate) manual drivers' average speed was below the speed limit, which could be explained by the experimental environment, because it is exhausting to drive at higher speeds for a longer period of time in the driving simulator.

However, this speed-reducing effect cannot be attributed to a potential role model function of Level 3 vehicle for manual drivers, as they maintained shorter time headways to lead vehicles on highway sections with higher CAV penetration rates compared to the sections with lower CAV penetration rates. This driving behavior resulted in close following and safety critical interactions with CAVs. Since these effects were evident in all three experimental groups, the safety criticality was not an effect of the eHMI, but a consequence of the CAVs' rule-following and defensive driving behavior. As discussed in the previous section (see section 4.2), this effect is attributable to the CAVs' behavior. Thus, manual drivers perceived the CAVs as obstacles rather than role models.

This perception was neither enhanced, nor mitigated by the external marking as the type of marking had no effect on manual drivers' ratings, or on their driving behavior. That is, knowing that the other vehicle is an CAV did not lead to a better anticipation of the behavior of this CAV, nor did it lead to participants adjusting their own driving style based on this knowledge. For example, a larger minimum distance would have been expected if participants had anticipated that the CAV would brake well in front of a speed sign. This was not even the case in the “correct marking” group, in which participants should have been able to develop adequate schemata about the CAVs' behavior throughout the simulator drive. Thus, the underlying issue does not appear to be that the manual drivers

were unable to understand and anticipate the driving behavior of CAVs. Rather, this defensive and slow driving behavior of CAVs frustrated manual drivers, causing them to maintaining (very) short minimum time headways to vehicles ahead.

However, in contrast to previous concerns about abusive driving behavior of manual drivers toward CAVs in mixed traffic (see e.g., Stanton et al., 2020), we found no systematic pattern of abusive behavior in the present study. From a methodology-driven perspective, it is important to note, however, that this issue might be difficult to examine in the driving simulator in general because participants know that they are being observed by the experimenter throughout the simulator drive. Therefore, participants may adhere to the rules to a greater degree than they would in their own vehicle in real-world driving.

After each highway section, manual drivers were asked how well they had progressed on the previous highway section. The results show that drivers had the (correct) impression of making significantly less progress when driving in mixed traffic than driving in non-automated traffic. In line with the results on perceived safety and comfort, this effect was already evident at a penetration rate of 25% CAVs (see Fig. 4, on the right). Despite the significant difference between non-automated and mixed traffic, ratings of perceived efficiency still equaled to the label *rather efficient* in mixed traffic. These findings indicate that manual drivers may have attempted to compensate for this lack of efficiency by keeping shorter safety margins to lead vehicles.

In summary, there is no evidence that CAVs act as role models for manual drivers and contribute to highway safety by reducing average speeds. Rather, the speed reduction appears to be due to CAVs acting as obstacles, leading to lower perceptions of safety and comfort, inducing human drivers to follow more closely and leading to a higher number of safety critical distances to vehicles ahead. As a result, a higher number of safety-critical interactions with vehicles ahead occurred. Although the defensive and slower driving of CAVs may increase safety for passengers in CAVs, surrounding manual drivers engaged in riskier driving behaviors and thus counteracted the positive speed-reducing safety effect of CAVs. This may particularly be the case when the penetration rate of CAVs in mixed traffic is low, as a small increase in safety from these vehicles may be outweighed by manual drivers engaging in unsafe distance keeping. Again, these findings may also be applicable to mixed traffic interactions with partially automated vehicles, as the assumptions made on CAV driving behavior and the relevant driving scenarios may also apply to vehicles travelling in Level 2 driving mode.

4.4. Potential benefits and issues of a status eHMIs on the highway

Regarding the use of a status eHMI on the highway, we found no significant differences between the “status eHMI” group and the other two experimental groups on manual drivers’ subjective ratings of perceived safety, comfort and perceived efficiency, or their driving behavior in terms of speed and distance keeping. Thus, the status eHMI had no effect on perceived safety, comfort, and driving behavior of manual drivers in terms of distance keeping, and thus no immediate benefit for traffic safety in the present study. This result is consistent with findings from previous research (see Fuest et al., 2020; Stange et al., 2022).

However, in the present study, manual drivers used the status eHMI to identify and distinguish CAVs from HDVs in mixed traffic, as shown by the results for the estimates of CAV penetration rates on the highway sections (see Fig. 5). This finding supports previous research (see Stanton et al., 2020), which hypothesized that a status eHMI as a visual cue could help human drivers to better anticipate and react (more appropriately) to CAVs in mixed traffic. Following this reasoning, one might speculate that a status eHMI might even accelerate the formation of adequate mental models compared to not using an eHMI. However, as discussed above, even in the “correct marking” group, where it can be assumed that drivers had sufficient time in mixed traffic to develop schemata of CAV driving behavior, the external marking had no effect on the subjective ratings or driving behavior.

Regarding the development of mental models, an alternative explanation for these results could be that manual drivers may not have needed to develop an entirely new mental model for CAV driving behavior. Instead, manual drivers needed to understand that CAV driving behavior is similar to that of a human driver with a defensive style. In other words, it might be possible that the CAV driving behavior we modeled in this driving simulator study was not completely outside the range of human driving behavior. Yet, manual drivers perceived this defensive driving behavior as less comfortable and less safe, and showed safety critical reactions when confronted with this type of defensive driving behavior. So, one may consider be some kind of abusive behavior toward CAVs or drivers with a defensive driving style (see Stanton et al., 2020), but this abusive behavior was not caused by external marking.

In summary, the results of the present study tend to discourage the introduction of eHMIs on the highway, as the effects of status eHMI on the outcome variables studied were not significant, except for the identification of CAVs in mixed traffic and their distinguishability from HDVs. These results suggest that the evaluation of automated driving behavior in mixed traffic by the manual drivers in our study was based less on the presence of an external marking and more on the driving behavior of these vehicles.

5. Limitations

The results obtained in the present study are subject to a number of limitations. First, participants had no personal experience with CAVs or driving in mixed traffic prior to this study, as these systems are not yet available on the market. In addition, it is likely that participants had different levels of knowledge or perceptions about how CAVs operate. Therefore, the information phase prior to the simulator drives was intended ensure that participants in the “status eHMI” and “no eHMI” groups shared a common level of knowledge. However, it remains unclear to what extent the informational phase, in which participants watched videos of automated (1) overtaking, (2) distance keeping, and (3) speed behaviors, sufficiently compensated for the lack of personal experience with CAVs and with driving in mixed traffic.

Second, the number of driving situations to be studied on the highway sections was limited by the number of driving situations that first-generation CAVs are likely to be able to handle independently. In these driving situations, we prototyped the CAVs’ driving

behavior to the assumed capabilities of a first-generation Level 3 vehicle, once the operational domain extends beyond congested traffic. According to the experts we interviewed, conditionally automated driving behavior may be subject to manufacturer-specific differences within the legal framework. So, the conditionally automated driving behavior implemented in the present study is an approximation to real automated driving behavior. In addition, human driving behavior is subject to a wide range, which could make the distinction between automated and manual driving behavior even more difficult in real-world driving (see [Stanton et al., 2020](#)) than in the present driving simulator study. It is possible that automated driving behavior is not completely outside the human domain, but is similar to a defensive human driver. Because we focused only on individual aspects of driving behavior in the present study, i.e., speed and distance behavior, the findings obtained are not fully transferable to real-world road traffic. Despite this limitation, the direction of the effects in the driving simulator and in real road traffic can be expected to be the same. Real world driving studies including prototypes of CAV and real manual drivers would thus be necessary to validate the results obtained in the present study. Nevertheless, our findings give a very good indication about the situations which should be examined in these studies. The present study should therefore be seen as a first step in exploring the perspective of manual drivers in mixed traffic beyond individual driving situations.

In the driving situations examined in the present study, we did not standardize the interactions between participants and the simulated vehicles, and participants had no further instructions regarding their driving behavior on the highway sections. Instead, participants should show their natural driving behavior in terms of general rule-following, lane-changing behavior, speed, and distance keeping. This may have resulted in participants experiencing different mixed traffic interactions with surrounding simulated vehicles during the simulator drives depending on their individual driving behavior. This lack of standardization may have affected the comparability of the simulator drives between participants to some extent. Nevertheless, the decision to standardize the road design of the highway sections instead of each individual interaction was advantageous in achieving greater realism and creating a more authentic experience of driving in mixed traffic for manual drivers. Regarding the driving behavior of manual drivers in mixed traffic and potential behavioral adaptations to driving in mixed traffic, future research needs to further investigate whether and to what extent manual drivers adjust their driving behavior in mixed traffic once they become more familiar with the presence of CAVs on the highway. This research question might require to a longitudinal study design to explore the familiarization process over a longer period of time.

In this context, we decided to exclude participants from the analysis of the questionnaire data because these participants already rated driving in manual traffic on the highway as *dangerous* or *unacceptable*. It remains unclear whether these ratings were due to driving on the highway or driving in the driving simulator. If the perceived safety scale in manual traffic on the highway has already been exhausted, it is hardly possible to measure change (ceiling effect). Therefore, the survey data in the present study are limited in their generalizability to the general driver population, as we only sampled participants who found highway driving in the driving simulator acceptable in the sense of uncomfortable or harmless.

Finally, the specific design of eHMIs for the highway use case is the subject of future research. In the present study, we used a turquoise light rectangle visible in a 360° radius around the simulated vehicle to mark the current automated driving mode (see [Stange et al., 2022](#); [Stanton et al., 2020](#)). However, research is still needed to determine whether light projection is technically feasible or what alternative medium, including LED light strips or displays, could be used and where the eHMI should ideally be placed (see [Bengler et al., 2020](#)), especially for the highway driving environment.

CRedit authorship contribution statement

Vanessa Stange: Conceptualization, Methodology, Software, Formal analysis, Data curation, Investigation, Visualization, Writing – original draft, Writing – review & editing, Project administration. **Matthias Kühn:** Supervision. **Mark Vollrath:** Conceptualization, Methodology, Software, Data curation, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

See [Table A1](#) and [Table A2](#).

Table A1

Multiple pairwise comparisons for the significant factor penetration rate on the questionnaire data.

		Penetration rate			
	Penetration rate	0%	25%	50%	75%
Perceived safety	0%	–	< .001	.077	.010
	25%	< .001	–	.150	.761
	50%	.077	.150	–	.221
	75%	.010	.761	.221	–
Comfort	0%	–	.028	.012	.023
	25%	.028	–	.590	.476
	50%	.012	.590	–	.694
	75%	.023	.476	.694	–
Perceived efficiency	0%	–	.017	.010	.005
	25%	.017	–	.724	.230
	50%	.010	.724	–	.214
	75%	.005	.230	.214	–
Perceived penetration rate*	0%	–	.766	.073	< .001
	25%	.766	–	.036	< .001
	50%	.073	.036	–	< .001
	75%	< .001	< .001	< .001	–

Note. Significant p-values in bold. *N = 25.

Table A2

Multiple pairwise comparisons for the significant factor penetration rate on the driving data.

			Penetration rate			
		Penetration rate	0%	25%	50%	75%
130 km/h	Average speed*	0%	–	.006	.040	.008
		25%	.006	–	.968	.108
		50%	.040	.968	–	.045
		75%	.008	.108	.045	–
	Minimum time headway***	0%	–	.641	.808	.180
		25%	.641	–	.924	.045
		50%	.808	.924	–	.167
		75%	.180	.045	.167	–
	Mean time headway***	0%	–	.507	.012	< .001
		25%	.507	–	.067	.001
		50%	.012	.067	–	.172
		75%	< .001	.001	.172	–
	Percentage of safety-critical interactions*	0%	–	.861	.015	.049
		25%	.861	–	.005	.048
		50%	.015	.005	–	.544
		75%	.049	.048	.544	–
100 km/h	Average speed*	0%	–	.005	.004	.007
		25%	.005	–	.616	.315
		50%	.004	.616	–	.375
		75%	.007	.315	.375	–
	Minimum time headway***	0%	–	.063	.963	.413
		25%	.063	–	.088	.312
		50%	.963	.088	–	.319
		75%	.413	.312	.319	–
	Mean time headway***	0%	–	.990	.677	.431
		25%	.990	–	.720	.345
		50%	.677	.720	–	.556
		75%	.431	.345	.556	–
	Percentage of safety-critical interactions*	0%	–	.694	.079	.024
		25%	.694	–	.088	.084
		50%	.079	.088	–	.247
		75%	.024	.084	.247	–

(continued on next page)

Table A2 (continued)

			Penetration rate				
			Penetration rate	0%	25%	50%	75%
80 km/h	Average speed*	0%	–	.100	.035	< .001	
		25%	.100	–	.493	.005	
		50%	.035	.493	–	.013	
		75%	< .001	.005	.013	–	
	Minimum time headway**	0%	–	.932	.415	.001	
		25%	.932	–	.168	< .001	
		50%	.415	.168	–	.005	
		75%	.001	< .001	.005	–	
	Mean time headway**	0%	–	.769	.245	.003	
		25%	.769	–	.041	< .001	
		50%	.245	.041	–	.015	
		75%	.003	< .001	.015	–	
	Percentage of safety–critical interactions*	0%	–	.556	.276	.113	
		25%	.556	–	.123	.047	
		50%	.276	.123	–	.601	
		75%	.113	.047	.601	–	

Note. Significant p-values in bold. *N = 49. ** N = 30. *** N = 27.

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